

Governing AI in Drug Discovery: AlphaFold3

Sung Song

Problem Statement:

- AlphaFold3: Neural Network Model based on deep learning algorithm
- Algorithm used to predict protein structures: has predicted over 200 million structures (previous version AlphaFold2)
- Faster drug candidate selection
- Needed in the drug industry: timeline for R&D is 10-15 years for approval
- Reduce cost of making one drug
- One drug cost over 2 billion dollars from conception to market approval

Ethical Framework:

- ❑ Bioethics is applicable here.
- ❑ Consequence-based theory: potential safety and efficacy issue for drugs
 - Morally not justifiable. This is just for the bottom-line for companies.
 - Not in line with utilitarianism for producing greatest good.
- ❑ Contract-based theory: potential unsafe drug in the market
 - Social contract theory: individuals have right to know this information
- ❑ Duty-based theory: participants in clinical trials are not informed properly of the risk of drugs made using AlphaFold3 (improper consent)
 - Humans should not be treated as means to an end. Autonomy not provided.
- ❑ Character-based theory: trade secret vs. public welfare
 - Virtue ethics: virtuous to protect the public from safety risks by providing information on potential unsafe drug from this technology.

Stakeholders:

- Pharmaceutical industry
- Health authority(ies)
- ICH (committee)
- Patients/consumers
- Healthcare professionals
- Health insurance carriers (commercial and public (Medicare/Medicaid) insurances)

Policy Solution:

- Push ICH guideline (guideline can be pushed globally)
- Regulations would follow after that
- Guideline: address ethical concerns that are safety related and regulatory as well
- Fill in the gap for GxPs for AlphaFold3
- FDA can have guidelines after that
- 21CFR would follow based on what can be codified
- Global health authority implementation following FDA as a model
- Pricing of drugs: legislative act

Objections/Critiques:

- ❑ Legislative act to control drug price:
 - Price of drug development expected to go down
 - No statistics on this at this point
 - If cost control for drug price not done right can impact drug industry's bottom-line
 - Can potentially impact innovation
 - Most drugs fail so the ones that make it have to make up for it with its profits
 - Suggest cost control when some data is available on how AlphaFold3 can lower R&D impact
- ❑ Improper informed consent process: need to balance risks/benefits ratio and communicated properly to subjects
 - Otherwise, can erode gains in speedy drug development and savings with challenges in retention/recruitment in research